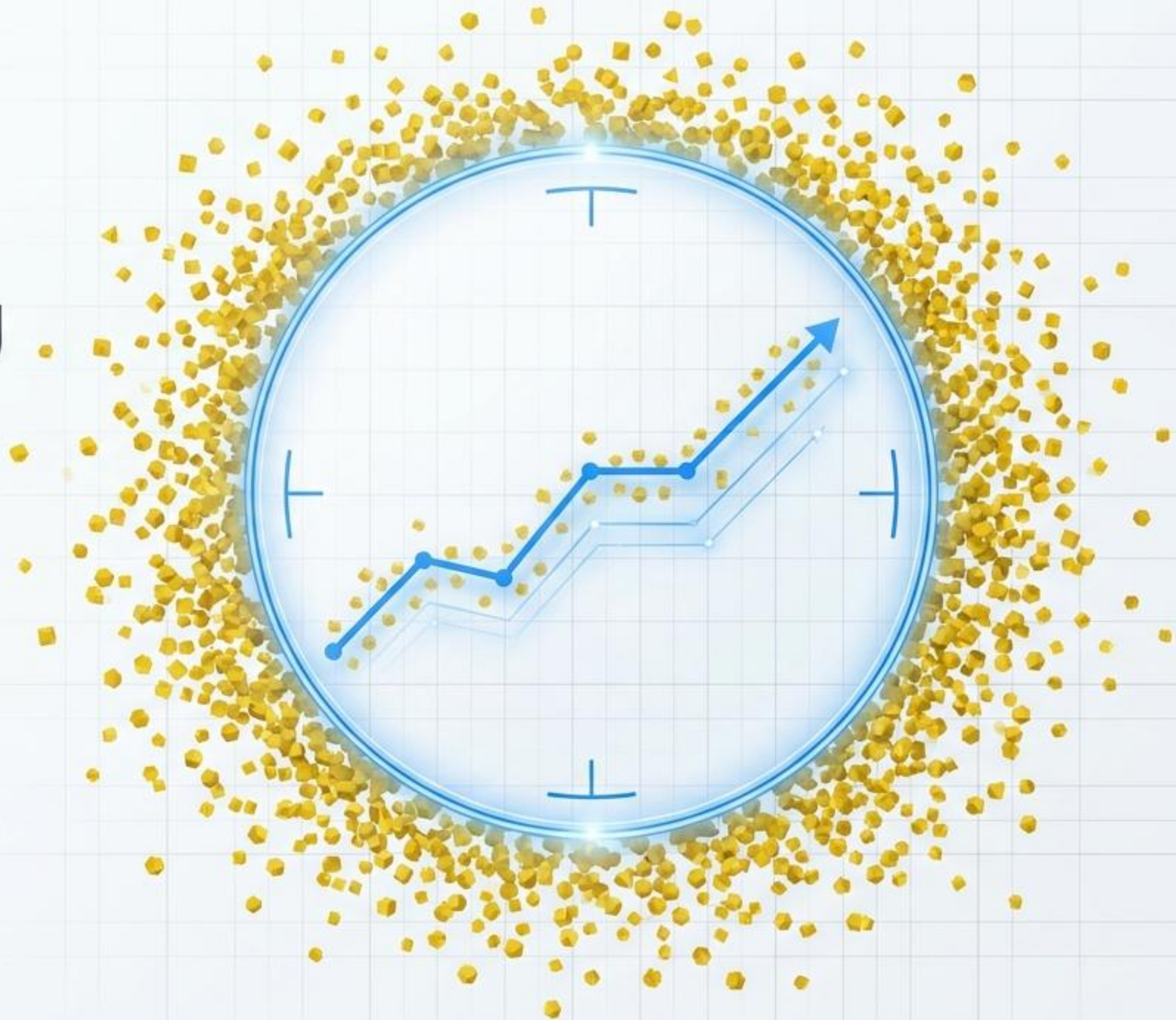


Power BI: Patterns and Trends Sensing

A comprehensive guide to uncovering hidden insights, anomalies, and drivers in your data model.

dr Zbigniew Galar



The 4 Pillars of Data Sensing

Trend Sensing



Revealing overarching direction.
Tools: Analytics Pane, Axis formatting

Anomaly Sensing



Isolating the exceptions.
Tools: Scatter Plots, Automated Variance Diagnosis

Structural Sensing



Forcing organization onto complexity.
Tools: Groups, Bins, Clusters

AI Sensing



Visualizing hidden statistical drivers.
Tools: Key Influencers, Decomposition Tree

Power BI is not just a visualization layer. It is an active diagnostic engine designed to surface insights the dominant data patterns hide.

Pillar 1: Trend Sensing (The Analytics Pane)

ANALYTICS PANE INTERFACE



Visual Dependency:
The Analytics feature is tied to the visual type. Line charts offer the maximum range of analytical overlays.

VISUAL OUTPUT & OVERLAYS

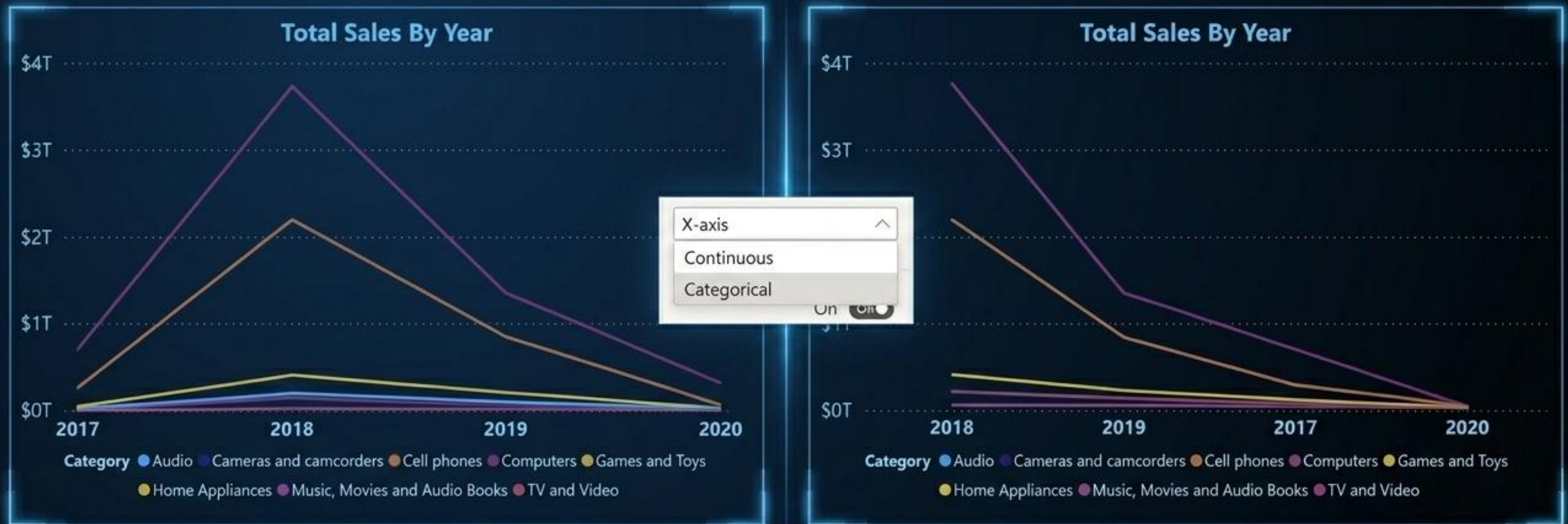


Available Overlays:
Trend lines, Constant lines, Average lines, Median lines, and (if supported) Forecasting.

Trend Sensing utilizes the Analytics Pane in Power BI to apply data-driven overlays onto visuals, quickly revealing the overarching direction of the data.

Time Series Foundations: Structural Angles

Toggle via Format visual > X-axis to radically shift the analytical perspective.

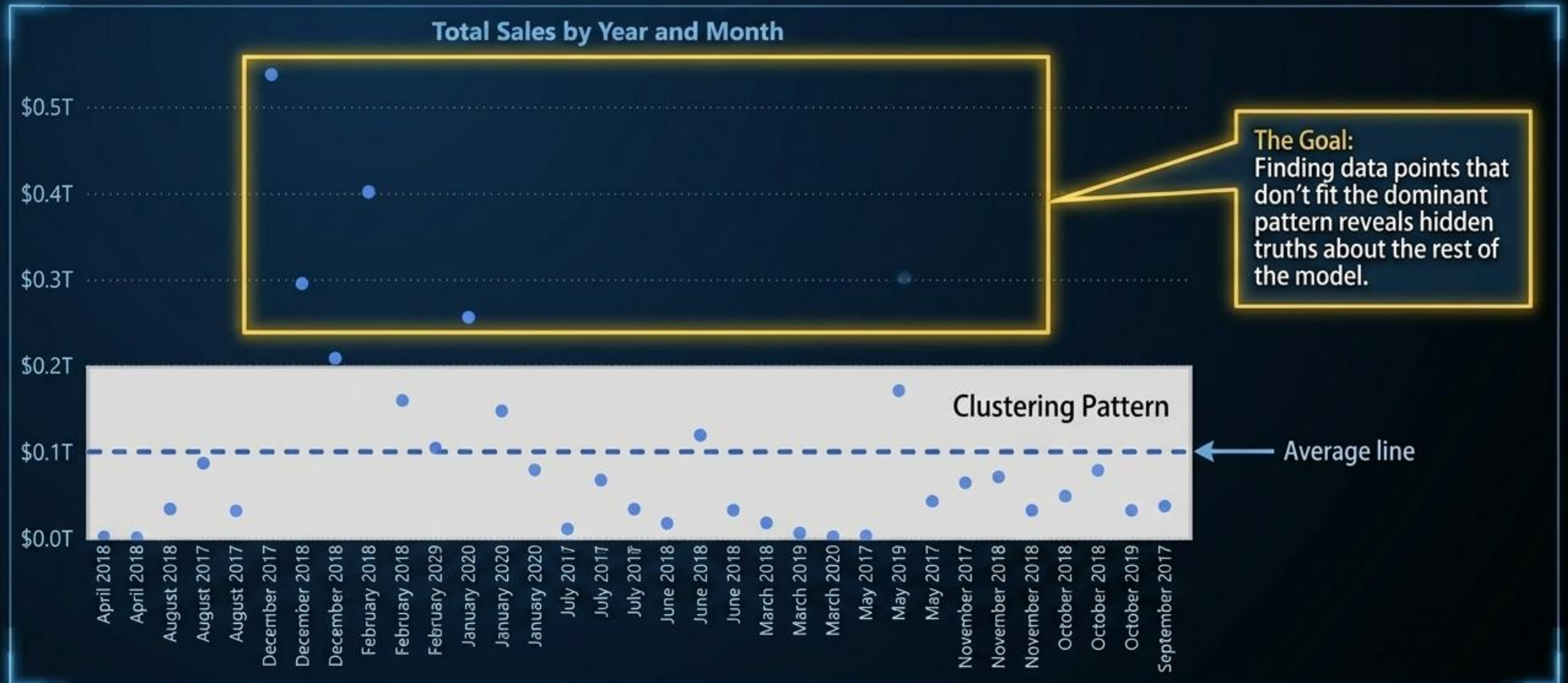


Continuous Axis: Data operates as an indivisible flow. It naturally plots chronologically to view true change over time.

Categorical Axis: Data operates as discrete points. Breaks chronological flow to allow sorting by value (e.g., highest to lowest performing periods).

Pillar 2: Anomaly Sensing (Identifying Outliers)

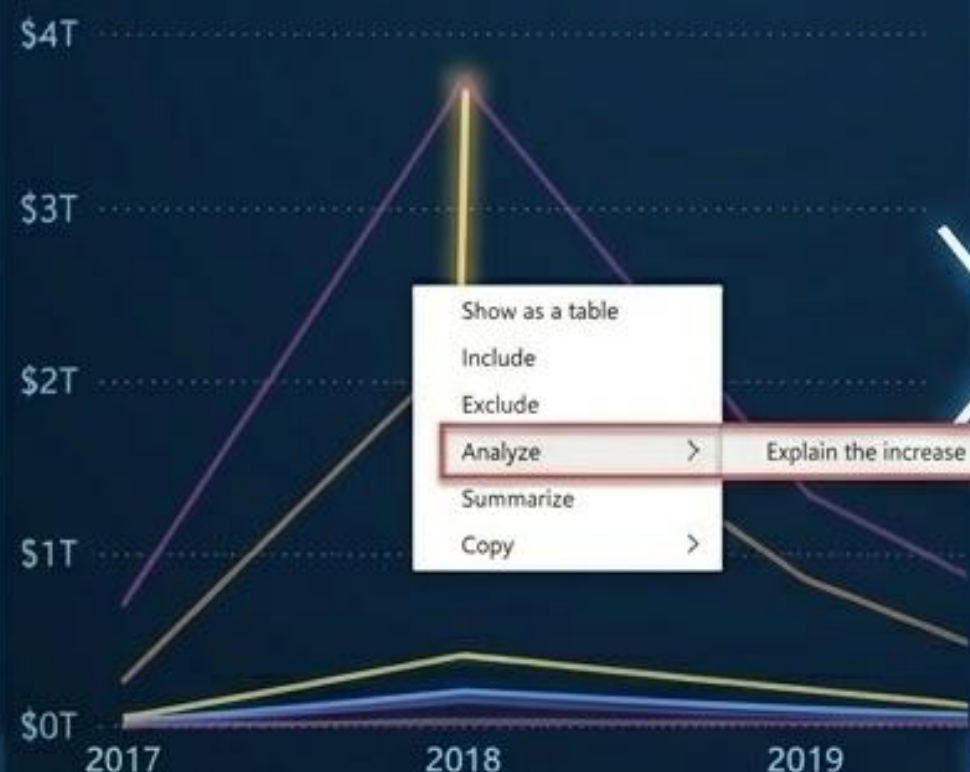
Combine Scatter Plots with Analytics overlays to force exceptions out of the noise.



Variance Diagnosis (The 'Analyze' Feature)

Automated, AI-driven root-cause detection.

1. Trigger



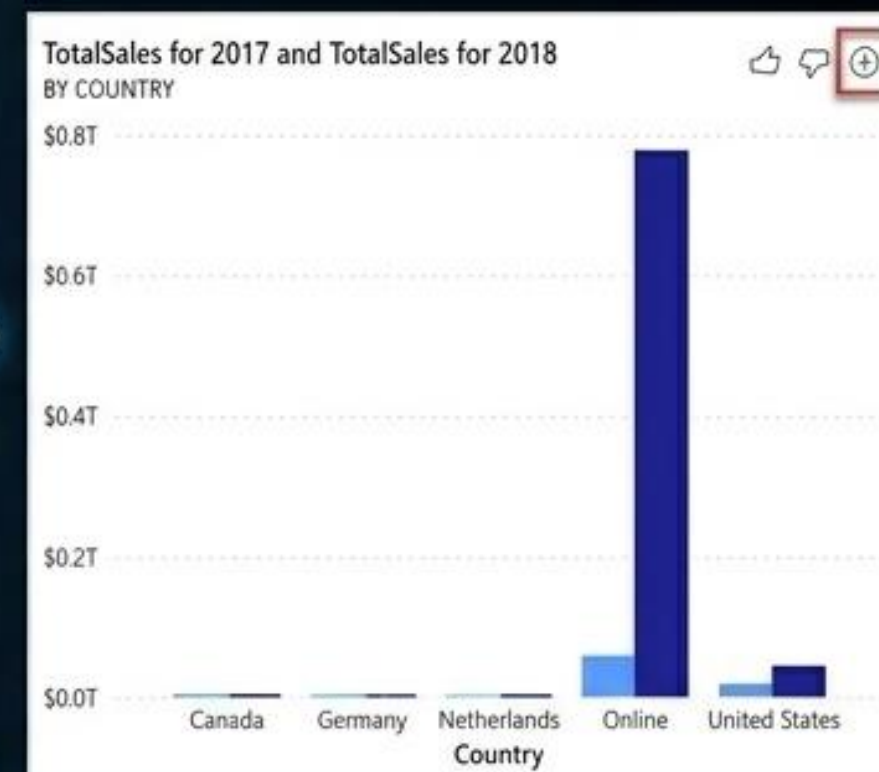
Right-click a dramatic change.
Power BI scans the entire
underlying data model.

2. Processing



The engine searches for
correlated variables across all
dimensions that drove the
specific variance.

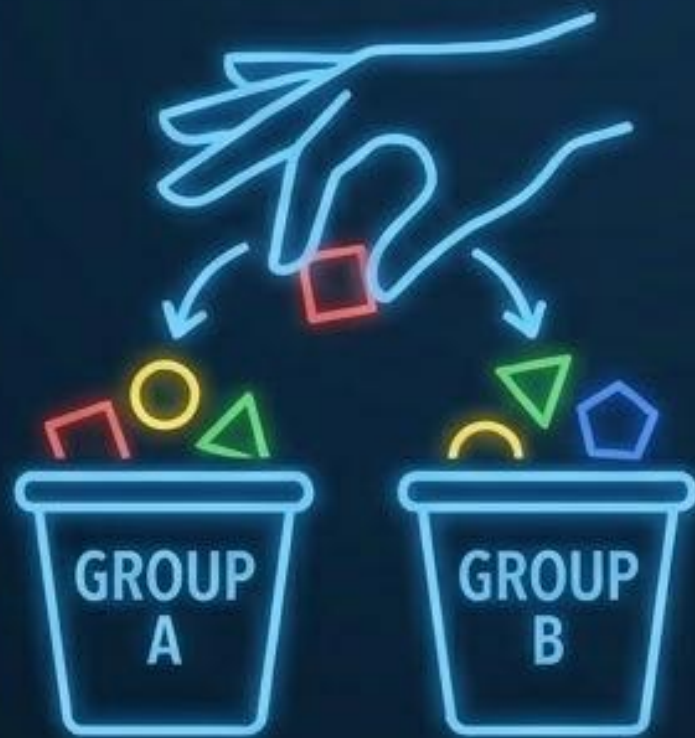
3. Result



Surfaces automated visual insights.
Click the '+' icon to instantly add
the chart to your report canvas.

Organizing Complexity: When Data Defies Analysis

Raw data often contains too many distinct values to analyze effectively. Power BI provides three distinct methodologies to force organization onto chaos, enabling macro-pattern recognition.



Groups

Handpicking specific discrete values.



Bins

Segmenting continuous flows into equal chunks.



Clustering

Letting algorithms discover hidden natural boundaries.

Structural Sensing: The Comparison Matrix

Groups	Bins	Clusters
Manual & Categorical	Rule-Based & Continuous	Algorithmic Machine Learning
Combining specific discrete values (e.g., merging individual states into custom 'Sales Territories').	Segmenting numeric or date flow into equally sized chunks (e.g., 10-year age brackets).	Letting statistical algorithms discover hidden natural segmentations based on multiple variables.
Works flawlessly with Measures.	Must be applied to Calculated Columns.	Applied directly to visual data points.

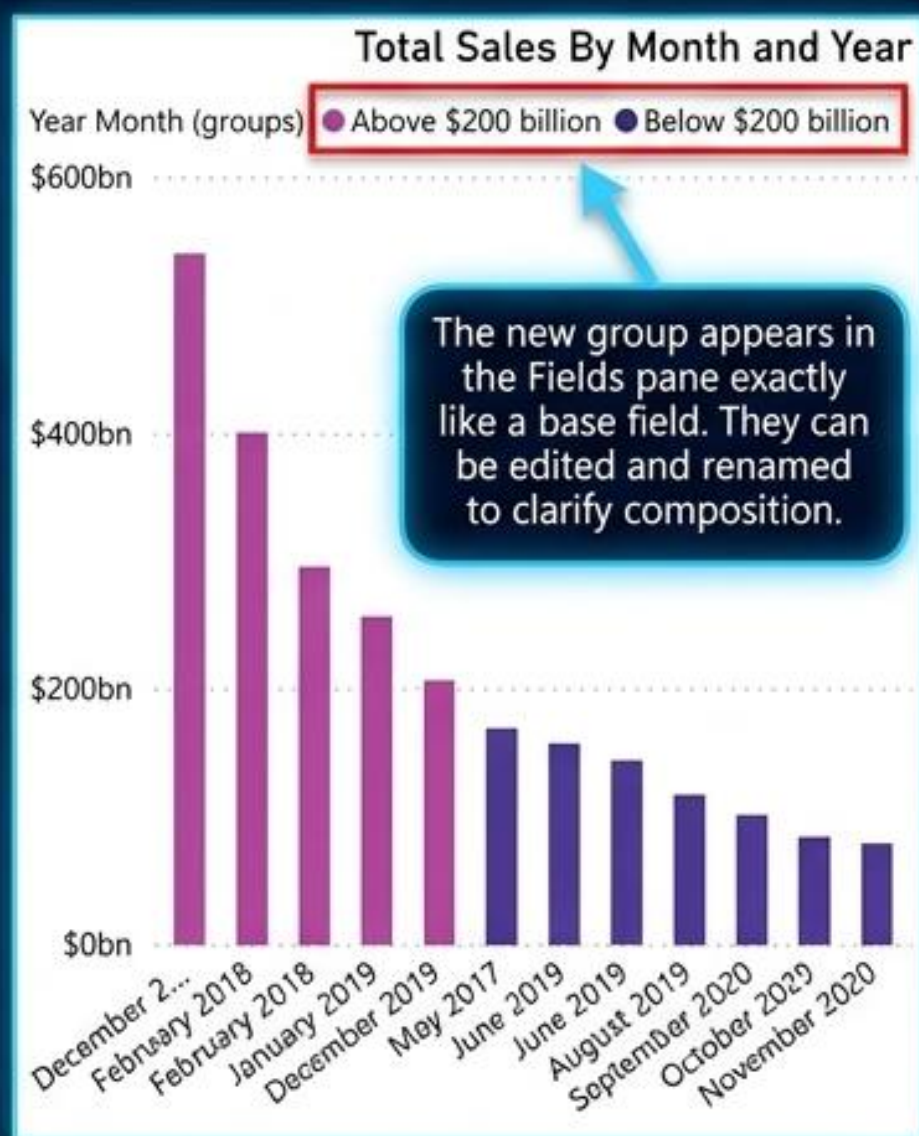
Deep Dive: Manual Groupings

Unlocking bulk data manipulation via custom categorical fields.

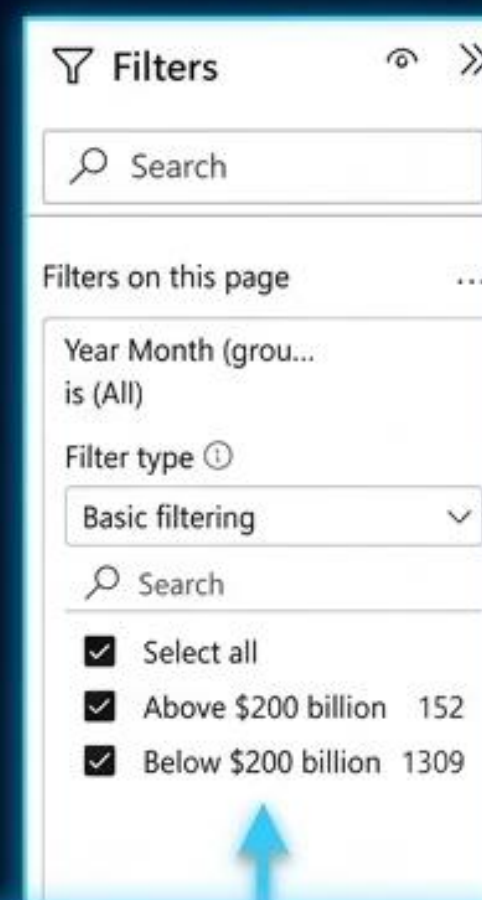
Creation



The Mechanics



The Power Multiplier



Because groups act as standard fields, they unlock macro-filtering. Drop them into Slicers and the Filters pane for instant bulk manipulation.

Deep Dive: Algorithmic Binning

Groups ×

Name *

Field

Group type

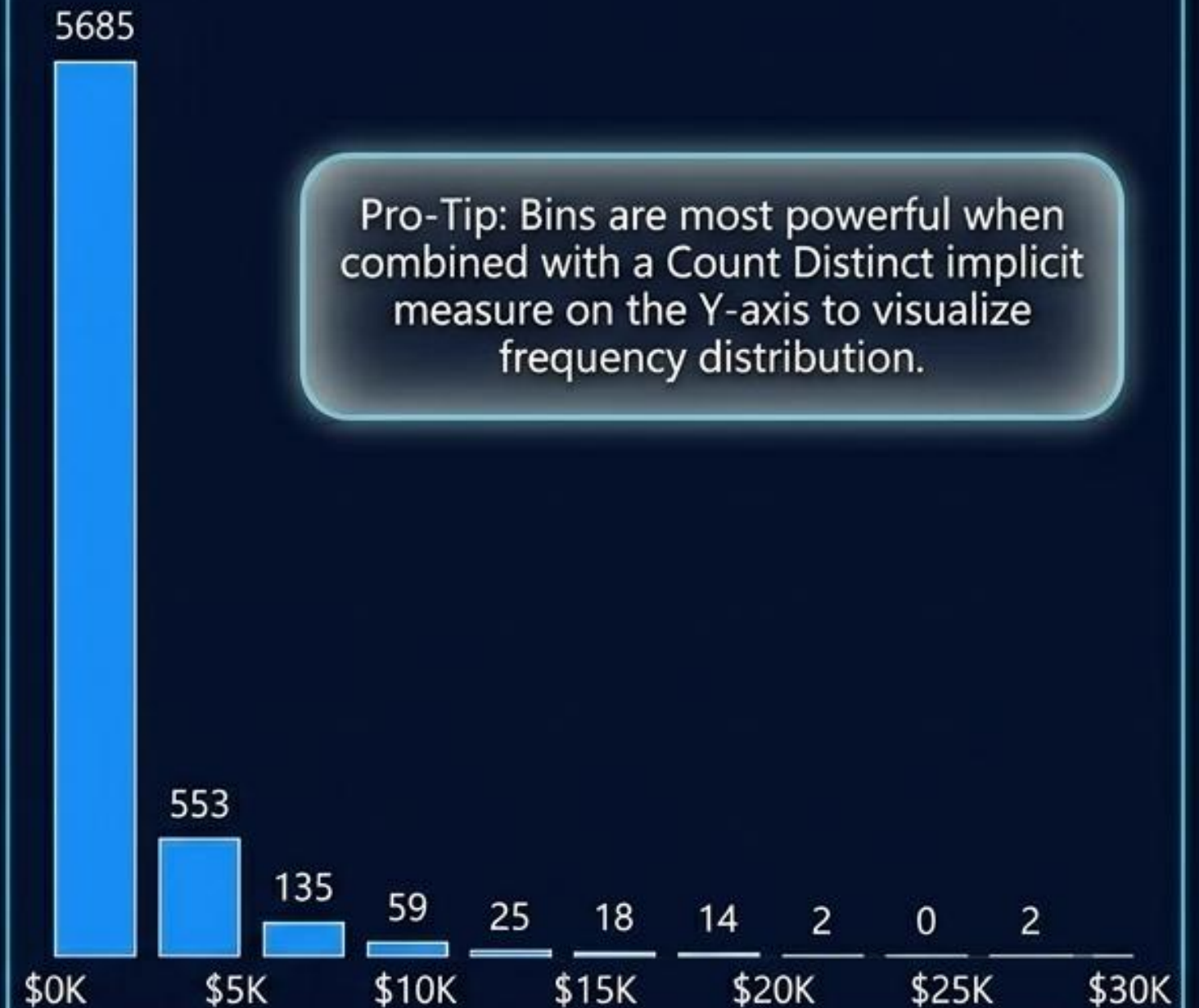
Bin type

Size of bins
(Fixed intervals,
e.g., every \$500)

Number of bins
(Fixed count, e.g.,
exactly 10 bins)

Right-click a Calculated Column
> New Group > Bin

Number of Orders in each Order Bin



Deep Dive: Statistical Clustering

Letting algorithms find natural boundaries humans cannot see.

The Discovery

TotalSales	Name	
202,752	Contoso Store Alaska	
210,923	Contoso Store Arkansas	
1193,226	Contoso Store Alars	
480,553	Contoso Store Aram	
206,978	Contoso Store Perbs	
300,147	Contoso Store Gormany	
299,883	Contoso Store Fumpery	
307,352	Contoso Store Floor	
93,764	Contoso Store Isyram	
653,935	Contoso Store Noxon	
408,865	Contoso Store Romam	
206,363	Contoso Store Store	
279,397	Contoso Store Alaska	

Click the ellipses menu on a multi-variable visual
> Automatically find clusters.

The Result

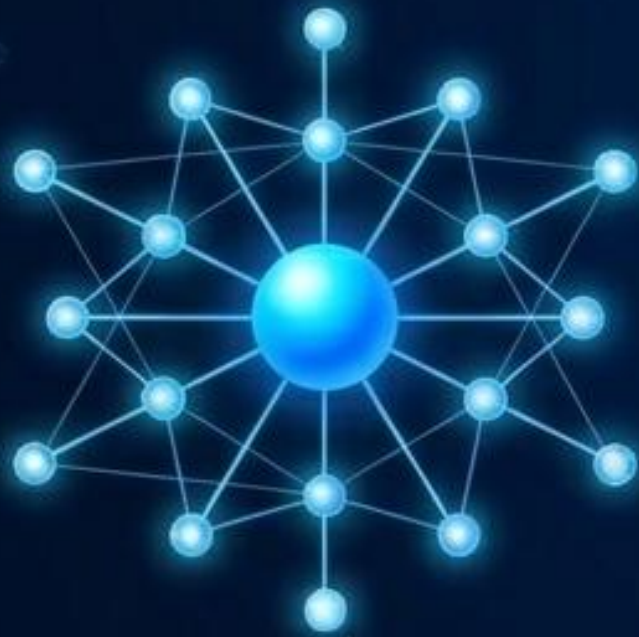
TotalSales	Name	Name (clusters)
202,752	Contoso Store Alaska	Cluster1
210,923	Contoso Store Arkansas	Cluster2
1199,226	Contoso Store Alars	Cluster3
480,583	Contoso Store Aram	Cluster1
206,978	Contoso Store Perbs	Cluster1
300,147	Contoso Store Clormany	Cluster2
299,883	Contoso Store Fumpery	Cluster3
307,352	Contoso Store Floor	Cluster1
93,744	Contoso Store Isrram	Cluster2
684,835	Contoso Store Nokon	Cluster3
408,366	Contoso Store Romam	Cluster3
204,383	Contoso Store Store	Cluster3
279,397	Contoso Store Alaska	Cluster3

The engine generates a completely new categorical field. This newly discovered segment can now be used as a primary slicing dimension across the entire report.

Pillar 4: AI-Powered Sensing

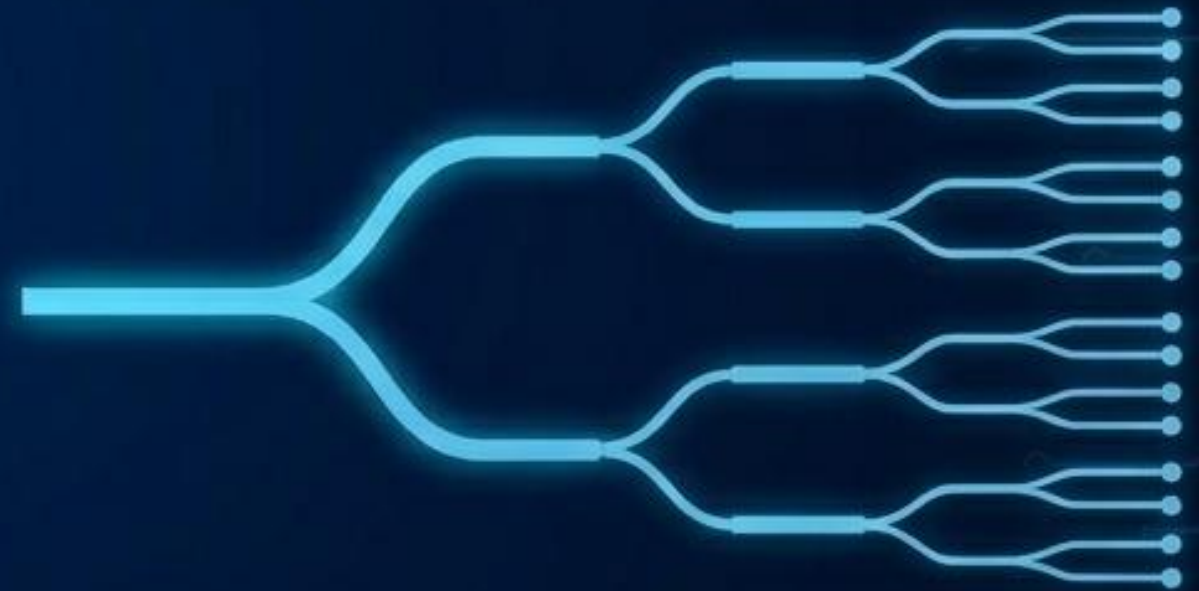
When facing millions of rows where patterns are completely obscured by sheer volume, traditional visual analysis fails. Power BI embeds specialized artificial intelligence to brute-force the underlying logic.

The Why Key Influencers



Focuses on isolating the exact statistical drivers and variables that cause a specific metric to increase or decrease.

The What Decomposition Tree



Focuses on elegant, multi-layered root-cause breakdowns, guiding the analyst through hierarchical dimensions.

Explaining the 'Why': Key Influencers

Uncovering the statistical drivers behind a specific metric.



Isolates variables
(e.g., Likelihood of
Total Sales increasing
multiplies by 5.89x
when category is
Music/Movies).

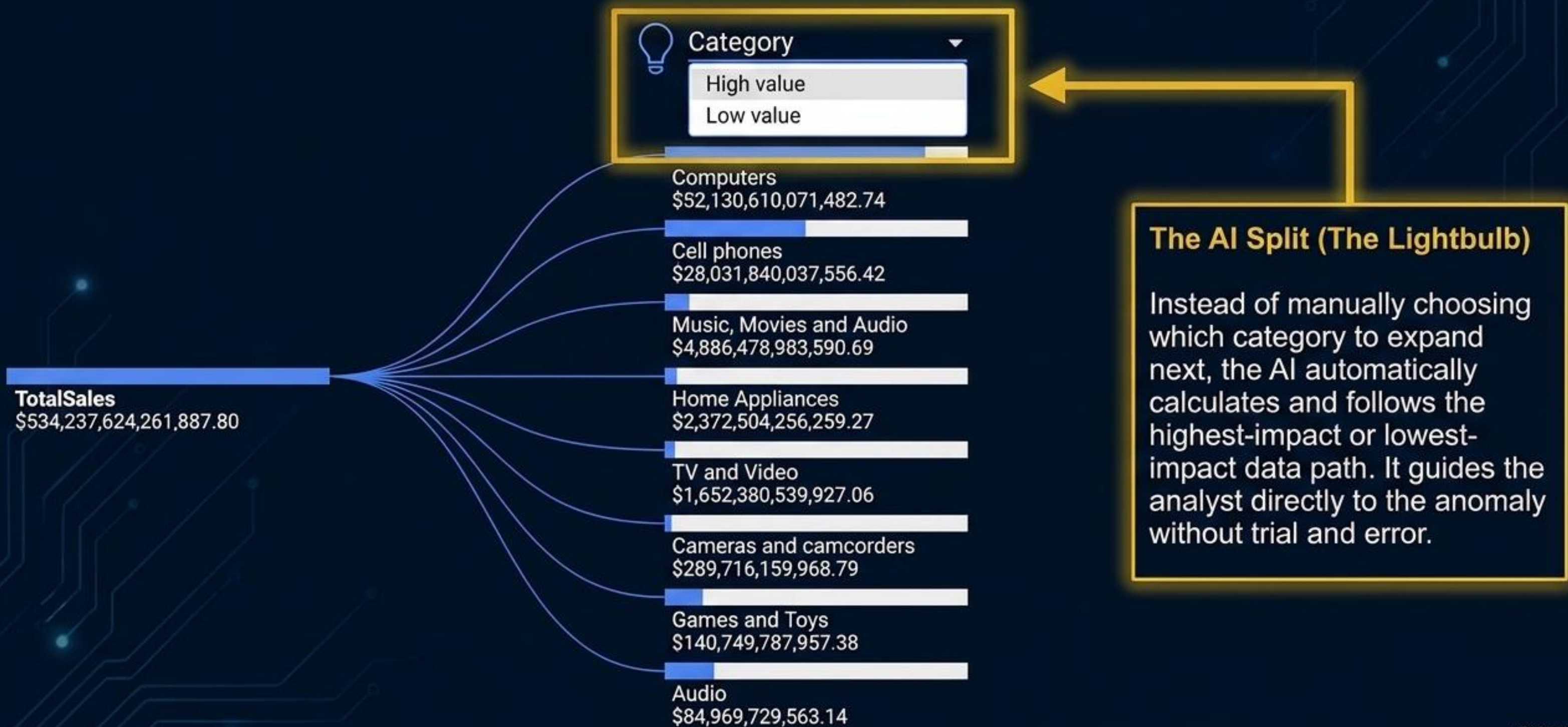
Secondary profiling
engine. Finds natural,
multi-variable
groupings in the data
that consistently drive
high or low values.

Crucial Setup:

Requires tuning in the Format visual pane. Analysts must dictate whether the Analysis Type is Categorical or Absolute.

Explaining the 'What': Decomposition Tree

Rapid, AI-assisted root-cause analysis through branching logic.



Synthesis: The Patterns & Trends Toolkit

What are you trying to sense?

Identify your analytical objective:

Need to see the
overarching
direction of noisy
data?

The Analytics Pane
(Trend/Average Lines)

Need to know *why*
a specific data
point spiked?

Automated Variance
Diagnosis
(Analyze > Explain
Increase)

Need to simplify
messy, discrete
categories
manually?

Groups
(Shift-click & bundle)

Need to track the
distribution of a
continuous
numeric flow?

Bins
(Size/Count rules)

Need to find
completely hidden
statistical drivers?

AI Sensing
(Key Influencers &
Decomposition Tree)

Mastery of Power BI is not just visualizing what *is*—it is sensing what is *hidden*.